EEE 498R

Direct Research Project Report



NORTH SOUTH UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

NAME OF PROJECT

MULTI-TASKING AUTONOMOUS DEVICE

Supervisor

Dihan Md Nuruddin Hasan

Assistant Professor, Department of Electrical and Computer Engineering

Course: EEE 498R | Section 5 | Group 13 | Summer 2023

Submitted By

Member	ID	Name
1	1911226043	Jubair Ibn Omar Hamza
2	1911622043	Sartaj Mahmud
3	1912325643	Shakil Khan
4	1911994043	Abdullah Al Mahfuz
5	1911075043	Nazifa Syed Aasfi

Date of Submission: _____

Agreement Form

We take great pleasure in submitting our Direct Research Project report on **"Multi-Tasking Autonomous Device"**. This report is prepared as a requirement of the Direct Research Project course EEE 498R, a one-semester design course in which groups of students build and test custom-designed systems, components or engineering processes. We request that you accept this report as partial fulfillment of the Bachelor of Science degree under the Department of Electrical and Computer Engineering, North South University.

Declared By:

.....
Name: Jubair Ibn Omar Hamza
ID: 1911226043

.....
Name: Sartaj Mahmud
ID: 1911622043

.....
Name: Shakil Khan
ID: 1912325643

.....
Name: Abdullah Al Mahfuz
ID: 1911994043

.....
Name: Nazifa Syed Aasfi
ID: 1911075043

Approved By:

.....
Supervisor
Dihan Md Nuruddin Hasan
Assistant Professor, Department of Electrical and Computer Engineering
North South University, Dhaka, Bangladesh

.....
Chair
Dr. Rezaul Bari
Department of Electrical and Computer Engineering
North South University, Dhaka, Bangladesh

Acknowledgement

We, the members of Group 13, are extremely grateful to our honorable supervisor, Assistant Professor Dihan Md Nuruddin Hasan, for his guidance, encouragement and constructive feedback throughout this project. We sincerely appreciate the time and support he gave us whenever we needed it.

We also thank the Department of Electrical and Computer Engineering at North South University for providing the academic environment and facilities that made this work possible, and our families and friends for their constant support during the semester.

Abstract

Robotics is one of the fastest-growing fields of engineering, and the COVID-19 pandemic has increased demand for devices that can clean and disinfect indoor spaces without exposing people to infection. This report presents the design, implementation and testing of a low-cost, Arduino-based **Multi-Tasking Autonomous Device** that combines several functions on one mobile platform: autonomous navigation with obstacle avoidance, ultraviolet (UV) LED disinfection that switches off automatically when human motion is detected, fire and gas hazard alarming, and a vacuum-cleaning module.

The device is built around an Arduino Uno and an L293D motor-driver shield driving DC motors. An HC-SR04 ultrasonic sensor mounted on a servo motor scans left and right whenever an obstacle is found within 30 cm, and the controller turns the vehicle towards the side with more free space. A PIR motion sensor turns the UV LEDs off whenever a person moves nearby, protecting skin and eyes from UV exposure. Temperature and gas sensors trigger a buzzer and warning LED when preset thresholds are exceeded. The design was verified through Tinkercad simulation and hardware prototyping; in our tests the obstacle-avoidance module prevented collisions in about 97% of trials. The prototype cost approximately BDT 3,000, and a full-featured production version is estimated to stay within BDT 5,000, making the concept affordable for homes, clinics and offices.

Keywords: autonomous robot, obstacle avoidance, ultrasonic sensor, PIR sensor, UV disinfection, fire and gas alarm, Arduino.

Table of Contents

Chapter 1: Introduction **1**
1.1 Project Details 1
1.2 Background and Motivation 2
1.3 Problem Statement 3
1.4 Project Objectives 3
1.5 Scope and Constraints 3
Chapter 2: Technical Details **4**
2.1 Proposed Solution 4
2.2 Solution Assessment & Project Goals 5
2.3 Design Alternatives 6
2.4 Technical Design: Module Level 7
2.5 Technical Design: System Level 7
2.6 Required Skills 7
Chapter 3: Essential Parts and Devices **8**
3.1 Technology Platform 8
3.2 Description of Components 8
3.3 Test Equipment and Hardware List 11
3.4 Work Breakdown Structure & Timeline 13
3.5 Financial Plan and Costs 14
Chapter 4: Methodology **16**
4.1 Related Work 16
4.2 Technical Approach 16
4.3 Circuit Design and Block Diagrams 21
4.4 Simulation 22
4.5 Coding and Implementation 24
Chapter 5: Project Summary **25**
5.1 Results and Discussion 25
5.2 Feasibility Study 26
5.3 Problems Faced and Troubleshooting 27
5.4 Limitations 27
5.5 Future Development 28
5.6 Project Demonstration Review 28
5.7 Lessons Learnt 29
5.8 Conclusion 29
Appendix A: Contribution of Individual Members **30**
Appendix B: References **31**
Appendix C: Codes **33**

List of Abbreviations

Abbreviation	Meaning
BDT	Bangladeshi Taka
CAD	Computer-Aided Design
DC	Direct Current
GSM	Global System for Mobile Communications
IDE	Integrated Development Environment
IR	Infrared
LED	Light-Emitting Diode
Li-ion	Lithium-ion (battery)
MDF	Medium-Density Fibreboard
PIR	Passive Infrared (motion sensor)
PWM	Pulse-Width Modulation
SMS	Short Message Service
UV / UV-C	Ultraviolet / ultraviolet radiation in the 200–280 nm germicidal band
WBS	Work Breakdown Structure

Chapter 1: Introduction

1.1 Project Details

Rapid advances in technology have brought a new era of automation in which machines can carry out tasks that were traditionally done by people. The multi-tasking autonomous device presented in this report is an example of this trend. It is a small, low-cost mobile robot that uses sensors, motors and a microcontroller to move around an indoor space on its own while performing several useful functions: avoiding obstacles, disinfecting surfaces with UV light, vacuum-cleaning the floor, and warning occupants of fire or gas hazards.

Robotics is a rapidly expanding field and one of the most direct ways to modernise everyday technology. We chose to work in this area to design something that makes daily life easier, especially in developing countries where affordable automation is still scarce. As a home or office appliance, the device offers a more convenient way to handle routine cleaning, while its safety features give users additional peace of mind by alerting them to hazards early.

Obstacle avoidance

The navigation system allows the device to travel through an unknown environment without colliding with objects. It detects an obstruction on its route, avoids it and continues moving. Common approaches to robot navigation include wall following, edge detection and line following. Edge-detection-based avoidance is widely used but has a drawback: the robot must stop in front of an obstacle to assess it accurately. Collision-avoidance algorithms range from simple ones that merely halt the robot when a barrier is detected to more advanced ones that let it detour around the obstacle; the latter require both obstacle recognition and some quantitative assessment of the obstacle's dimensions. Our device takes a middle path: it stops, scans both sides with a servo-mounted ultrasonic sensor, and turns towards the side with more free space.

UV disinfection

The UV disinfection module is intended to sanitise indoor spaces without chemical disinfectants or direct human involvement. UV-C radiation (200–280 nm) is well known for its germicidal effect on bacteria and viruses, and is used in hospital disinfection robots [1], [2], [8]. Because UV radiation is harmful to skin and eyes, the module is coupled with a PIR motion sensor so that the UV LEDs switch off whenever a person moves nearby.

Fire and gas alarm

The alarm module monitors the surroundings for signs of fire or a gas leak. A temperature sensor and a gas sensor feed the Arduino, which compares the readings with preset thresholds and, when a threshold is exceeded, activates a buzzer and warning LED to alert the occupants. The same principle can be extended with GSM-based SMS notification [13] and, in a future version, with automatic shut-off or suppression mechanisms.

Vacuum cleaning

The vacuum module uses a small DC motor with a fan and suction opening to draw dust and debris into a collection chamber. The motor is switched by the controller, while the ultrasonic sensors steer the device around obstacles so that it can cover the floor area automatically [11], [12].

1.2 Background and Motivation

The rapid spread of the highly infectious COVID-19 virus caused far-reaching social and economic changes and forced people to look beyond social distancing and government guidelines for ways to reduce the spread of infectious pathogens. Sanitation and sterilisation have since become an inseparable part of daily life, and the swift growth of a germ-conscious mindset among the general public has accelerated interest in automated cleaning, sanitising and disinfecting devices. However, with communicable diseases caused by airborne microbes, it is unwise for people to take part physically in the sterilisation process, because of the considerable risk of contracting the disease from a contaminated space.

Necessity is the mother of invention, and this situation calls for a versatile, efficient and autonomous device that can sterilise a space with relative ease and without direct human involvement. Robots are already used in several applications where people would be exposed to risk. We therefore set out to build a UV disinfection module that can be used in hospitals as well as in apartments and offices.

Fire and air quality are further concerns. A study of burn epidemiology in Bangladesh reports that nearly 3,000 people die each year from burn-related injuries; fire alarm systems play an important role in warning occupants and thereby help save lives. Poor indoor air is another major health burden: indoor air pollution is estimated to cause about 3.8 million deaths each year, since particulate matter and harmful gases can lead to asthma, reduced lung function and even cancer. Leaks of combustible or toxic gases such as LPG, methane, H_2S or SO_2 can cause explosions, poisoning and oxygen displacement, so early detection allows occupants to evacuate and limits damage.

The vacuum-cleaning module adds versatility, making the device attractive to people who want help with household cleaning. It sweeps across the floor, sucks up dirt on the way and can be stopped at will.

The novelty of this project is that it combines these functions in one compact, robust and low-cost robot-like device. The device can avoid obstacles by sweeping an ultrasonic sensor left and right (a simple radar-like scan), switch off its UV LEDs when human motion is detected nearby, disinfect an enclosed space without external intervention, vacuum the floor, and raise an alarm when it senses heat or gas. Its uniqueness lies in this multi-functionality at a price that ordinary households and small institutions can afford.

1.3 Problem Statement

Cleaning and maintaining indoor spaces is time-consuming and labour-intensive, and it becomes dangerous when the space must also be disinfected. At the same time, hazards such as fire or gas leaks can go unnoticed and require quick, decisive action. Existing cleaning and safety technologies address these needs only separately: robotic vacuum cleaners do not disinfect or monitor hazards, UV disinfection robots are expensive and rarely clean floors, and fire or gas detectors are fixed devices that cannot move through a space.

The multi-tasking autonomous device aims to close this gap by providing one affordable platform that navigates on its own, cleans and disinfects, and warns occupants of hazards. Automating these tasks frees up time and resources, and the safety features help occupants react quickly in an emergency.

Several challenges must be addressed to achieve this: integrating multiple sensors and control tasks on a limited microcontroller, supplying enough power for motors, UV LEDs and the vacuum motor from a small battery, and ensuring that the device operates safely and reliably in different indoor environments. Meeting them requires careful design, simulation, testing and iteration.

1.4 Project Objectives

The objectives of the project are to:

- design and build an Arduino-based mobile platform that navigates an unknown indoor space and avoids obstacles by scanning left and right with an ultrasonic sensor;
- implement UV-LED disinfection that switches off automatically when a PIR sensor detects human motion nearby;
- monitor temperature and gas level and raise an audible and visual alarm when preset thresholds are exceeded;
- integrate a vacuum-cleaning module for automatic floor cleaning;
- verify the design through Tinkercad simulation and hardware testing; and
- keep the cost low, with a prototype budget of about BDT 3,000 and a target of at most BDT 5,000 for a full-featured production version.

1.5 Scope and Constraints

The prototype is designed for flat indoor floors in homes, offices, clinics and similar spaces. It is a proof of concept built from off-the-shelf hobby-grade components and powered by a small rechargeable battery. Three constraints shape the work: PIR sensors respond to motion and may miss a person who stands perfectly still; the disinfection effectiveness of the UV LEDs depends on wavelength, distance and exposure time and was not measured quantitatively; and the run time is limited by battery capacity. These points are discussed further in Chapter 5.

Chapter 2: Technical Details

2.1 Proposed Solution

To address the challenges of cleaning indoor spaces and detecting potential hazards, we propose a multi-tasking autonomous device. In our system a single ultrasonic sensor, mounted on a servo motor, sweeps left and right and the controller uses a predefined algorithm to decide where the vehicle should move next. We tried to make the system as cost-effective as possible. The initial version has a front-facing sensor to demonstrate the concept; rear and side sensors can be added according to a client's requirements. We preferred the ultrasonic sensor over alternatives because it gives reliable distance data in unfamiliar environments and works regardless of lighting. If faster or more accurate processing is required, and cost permits, the Arduino can be replaced by a Raspberry Pi or a custom microcontroller; initially we use the Arduino.

The proposed device is a compact and portable autonomous platform that avoids obstacles, switches off its UV LEDs when human motion is detected nearby, disinfects an enclosed space without human intervention, vacuum-cleans the floor, and sounds an alarm when it detects fire-related heat or gas. It consists of the following main elements:

- **Microcontroller:** an Arduino board receives and processes input from the sensors and controls the motors and other components.
- **Sensors:** an ultrasonic sensor detects objects in the path, a PIR sensor detects human motion, and temperature and gas sensors detect fire or leak hazards.
- **Motors:** DC motors drive the wheels and the vacuum fan, and a servo motor rotates the ultrasonic sensor.
- **Power supply:** a rechargeable battery supplies the motors and electronics, with regulation to 5 V for the logic circuits.
- **Control system:** the Arduino program lets the device navigate, avoid obstacles, control the UV LEDs and raise alarms.

By combining these elements in a single device, the system offers a more efficient way to clean indoor spaces while also enhancing safety by detecting hazards and alerting occupants.

2.2 Solution Assessment & Project Goals

The key requirement of the project is the safety of the device itself: it must detect an obstacle, stop its acceleration and brake, and then decide where to move next. We assessed the solution design in terms of alignment and quality. The system was installed on a demonstration toy car with no dedicated braking system. It synced well with the car and worked properly: when an obstacle appears within the predefined range, the car stops accelerating and then turns left or

right depending on which side has the larger distance. This protects the device from collisions, and the project meets the requirements of the stakeholders and the proposed solution.



Figure 1: Obstacle-avoidance concept: the ultrasonic sensor detects an obstacle and the vehicle turns towards the clearer side.

The overall goals are summarised by the functional modules below.



Figure 2: Functional modules: UV sterilisation, fire alarm system and vacuum cleaning.

A) **Sterilisation:** The system is designed to sterilise an enclosed place. Equipped with UV lights, it is intended to inactivate infectious microbes and pathogens such as viruses, bacteria, fungi and spores on exposed surfaces. Disinfection effectiveness depends on the UV wavelength, intensity, distance and exposure time.

B) **No human intervention:** The device moves across the floor of a closed space without human intervention, reducing the risk of contracting communicable pathogens. Autonomous technology enriches automated systems with sensors, artificial intelligence and analytics so that they can make independent decisions from the data they collect.

C) **Obstacle avoidance:** The device detects obstacles automatically and redirects its path to avoid collisions. In robotics, obstacle avoidance means satisfying a control objective subject to non-collision position constraints. IR sensors are widely used for distance measurement and can also be used for this purpose; however, although IR sensors respond faster than ultrasonic sensors, ultrasonic sensors are the better choice for small, relatively slow devices like ours.

D) **Switch-off near human motion:** The UV lights are switched off automatically when a person moves nearby, to avoid exposing human skin and eyes to UV radiation. A PIR sensor detects human presence from the change in infrared radiation across the sensor's detection zones, so it cannot reliably detect a person who is completely stationary (a

false negative). Human motion in close proximity is, however, detected clearly, which is enough to switch the UV LEDs off.

E) **Vacuum cleaning:** A motor-driven suction system collects dust and debris through a suction opening and carries it into a storage container inside the device. Attachments can be used to clean different surfaces such as carpet, hardwood or tile.

F) **Fire and gas alarm:** When the temperature sensor or gas sensor reading crosses its threshold, the controller activates an alarm, giving a loud buzzer tone and a warning LED so that occupants are alerted.

2.3 Design Alternatives

Three design alternatives were considered before the final design was chosen (Table 1). The first is to use a Raspberry Pi instead of the Arduino. Both can run the same logic, but the Raspberry Pi has far more processing power and therefore responds more smoothly; it also allows a larger power supply unit to be used and the weight to be balanced accordingly, though at higher cost and power consumption. The second is to drop the servo motor and moving head and instead mount two additional sonar sensors on the left and right sides. In the code, the side sensors would be read only when the front sensor detects a nearby obstacle, and the longer obstacle-free side would be chosen. The third is to use IR sensors in place of ultrasonic sensors.

Table 1: Design alternatives considered

Alternative	Advantage	Drawback
Raspberry Pi instead of Arduino Uno	More processing power; smoother response	Higher cost and power consumption; more complex to set up
Two extra side sonar sensors instead of a servo-mounted sensor	No moving part; both sides measured at once	More sensors (cost and wiring); extra code logic
IR sensors instead of ultrasonic	Faster response time	Less suitable for small, slow devices; ultrasonic preferred for reliable range data

2.4 Technical Design: Module Level

At the module level the system is highly modular: each component has its own function, and combining them makes the system work. The design is highly cohesive. Modules such as the sonar sensor, the wheels and the servo motor show **functional cohesion**, because every element needed for a single computation is contained in the component.

The whole system also shows **sequential cohesion**, in which the output of one element becomes the input of the next. The motors run when the microcontroller sends the signal and

stop when the sonar sensor detects an obstacle; the servo motor runs only when there is an obstacle, to measure the right and left distances with the sonar sensor. The wheels operate on the same input data, which is an example of **communicational cohesion**. There is no sign of coincidental cohesion.

The modules are loosely coupled. They are mostly connected by **data coupling**, in which components are independent and communicate only through data, and by **external coupling**, in which modules depend on hardware or software outside the program being developed. There is no **content coupling**, so one module cannot modify the data of another.

2.5 Technical Design: System Level

At the system level, the design describes how data is entered, verified, processed and acted upon as output. The sonar sensor continuously supplies data (the measured distance, a number) to the microcontroller, which processes it and outputs commands: for example, a Boolean decision on whether the motors and the servo motor should run. The PIR and gas/temperature sensors feed additional inputs, and their outputs control the UV LEDs, the buzzer and the warning LED. The system is sequential: each stage depends on the result of the previous one.

2.6 Required Skills

The skills required to build the system are modest. A clear understanding of the requirements and the expected output is essential. The control algorithm was written in the Arduino language, which is based on C++; Python or C could have been used as well. We learnt how the microcontroller is wired and how its functions work in order to establish the connections, and our basic knowledge of electronics made this easy to pick up. Familiarity with circuit simulation (Tinkercad) was also helpful for testing before assembly.

Chapter 3: Essential Parts and Devices

3.1 Technology Platform

The technology platform of the project consists of hardware and software components.

Hardware: the main components are the microcontroller board, the sensors (ultrasonic, PIR, temperature and gas), DC motors with a motor driver, a servo motor, the vacuum-cleaner mechanism (suction opening and motor), UV LEDs, a rechargeable battery and the mechanical parts required for movement.

Software: the microcontroller is programmed in the Arduino Integrated Development Environment (IDE). The program interprets the sensor signals and controls the other components. Additional tools were used for design and simulation: Tinkercad for circuit simulation, Multisim for circuit analysis, MATLAB for computation, and CAD software for the mechanical parts where needed.

Overall, the platform integrates these hardware and software parts into an automated system able to perform several tasks without human intervention.

3.2 Description of Components

Arduino Uno

The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P and developed by Arduino.cc. It has 14 digital input/output pins, of which 6 can be used as PWM outputs, and 6 analog inputs. It also has a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button. It is widely used in electronic projects because it is low-cost, flexible and easy to program.



Figure 3: Arduino Uno board.

Motor driver

A motor driver is an interface between the motors and the control circuit. The microcontroller cannot supply the current that motors need, so the driver takes low-power control signals and switches the motor supply accordingly, controlling speed and direction. L293D and ULN2003 are two of the most commonly used motor-driver ICs. The L293D contains two H-bridges and

can drive two DC motors in both directions (up to about 600 mA per channel). In this project the L293D motor-driver shield is used together with the AFMotor library to drive the DC motors and the servo.



Figure 4: L293D motor driver.

Connecting wires (jumpers)

Jumper wires have connector pins at each end, so they can connect two points without soldering. They are typically used with breadboards and other prototyping tools so that a circuit can be changed easily; the colours of the wires carry no meaning. Jumper wires come in male-to-male, male-to-female and female-to-female versions; the difference lies in the end points of the wire. Male ends have a protruding pin and plug into things, while female ends accept a pin.



Figure 5: Jumper wires.

Double-sided tape

Double-sided tape is a pressure-sensitive tape coated with adhesive on both sides, used to stick two surfaces together. In this project it was used to fix the lightweight modules and sensors to the chassis.



Figure 6: Double-sided tape.

Battery holder

A battery holder is a compartment that holds one or more batteries and makes electrical contact with their terminals. The holder used here connects four AA cells in series to power the project.



Figure 7: Battery holder.

Sonar (ultrasonic) sensor

An ultrasonic (sonar) sensor measures distance by sending out a short burst of ultrasound and timing the echo that returns from an object. In this project we use the HC-SR04 module, which measures distances from 2 cm to 400 cm with an accuracy of up to 3 mm and a measuring angle of approximately 15°. It has four pins: VCC, Trig, Echo and GND.



Figure 8: HC-SR04 ultrasonic sensor.

Servo motor

A servo motor is a rotary or linear actuator that allows precise control of angular or linear position, velocity and acceleration. It consists of a motor coupled to a sensor for position feedback. In this project a small SG90 servo rotates the ultrasonic sensor to the left, right and centre.



Figure 9: SG90 mini servo motor.

Software tools

- **Arduino IDE** contains a text editor, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them.
- **MATLAB** is a high-performance language for technical computing that integrates computation, visualisation and programming in an easy-to-use environment.
- **Multisim** integrates industry-standard SPICE simulation with an interactive schematic environment to visualise and analyse the behaviour of electronic circuits.
- **Tinkercad** is a browser-based tool for simulating Arduino circuits and code, and was used to verify the PIR and gas/temperature circuits before assembly.

3.3 Test Equipment and Hardware List

Table 2 summarises the purpose of each main component, and Table 3 lists the hardware used in the prototype.

Table 2: Purpose of the main components

Component	Purpose in this project
Arduino Uno	Main controller. It can be connected to a computer by USB for powering and programming, and it runs the program written in the Arduino language.
Motor driver (L293D)	Interface between the motors and the control circuit. The Arduino outputs only 5 V at a few milliamps, so a driver is needed to switch the higher motor supply.
Connecting wires (jumpers)	Connect two points of the circuit without soldering.
Sonar sensor (HC-SR04)	Measures distance and detects obstacles. One pin (Trig) transmits the ultrasonic pulse and another (Echo) receives the reflection.

Component	Purpose in this project
Servo motor (SG90)	Rotates the ultrasonic sensor left and right so the distance on each side can be compared.
DC motors	Drive the wheels and move the vehicle.
PIR sensor (HC-SR501)	Detects human motion nearby so the UV LEDs can be switched off.
UV LEDs	Provide the ultraviolet light used for disinfection.
Temperature and gas sensors, buzzer	Detect heat and gas hazards and sound the alarm.
Battery	Supplies power to the motors and electronics; a rechargeable pack allows repeated use.
Wheels	Reduce friction and allow the vehicle to move and turn.
Chassis	The main support structure (frame) that carries the components and bears the mechanical stresses.

Software used: (a) MATLAB, (b) Multisim, (c) Arduino IDE for Windows, (d) Tinkercad.

Table 3: Hardware list

No.	Component	Image
1	HC-SR04 ultrasonic module	
2	L293D motor driver	
3	5 V N20 motors and mounting brackets	
4	N20 motor wheels	
5	UV LEDs	
6	7.4 V Li-ion battery	
7	7805 voltage regulator	
8	330 Ω resistor	
9	Breadboard	

No.	Component	Image
10	Polyester (perf) board	
11	Castor wheel	
12	Arduino Pro Mini (ATmega328P, 5 V / 16 MHz)	
13	PIR sensor HC-SR501	

3.4 Work Breakdown Structure & Timeline

The work was divided into five modules:

A) **Autonomous obstacle-avoiding vehicle:** detects obstructions in its path and avoids them. The Arduino is the central control unit that processes data and sensor signals and sets the parameters that guide the vehicle. The module is fully automatic and needs no human control when placed in an unfamiliar environment.

B) **UV-LED disinfection:** the UV LEDs sterilise an enclosed place. The module is designed to deactivate microorganisms such as viruses, bacteria, fungi and spores on exposed surfaces.

C) **Human-presence cut-off:** a PIR sensor switches the UV LEDs off when human motion is detected nearby, so that skin and eyes are not exposed. Because PIR sensors respond to motion, a perfectly still person may not be detected; nearby human motion, however, is detected clearly and is sufficient to switch the LEDs off.

D) **Fire and gas alarm:** the temperature and gas sensors send analog signals to the Arduino. When a reading exceeds its threshold, the Arduino rings a buzzer and lights a warning LED; it could also trigger a siren or a fire-suppression action.

E) **Vacuum cleaning:** a suction motor and fan collect dust and debris and transport them to a storage container inside the device.

Timeline

Table 4: Project timeline

Week	Activity
Week 1	Research and literature review
Week 2	Circuit design and flow chart
Week 3	Programming, simulation and data extraction
Week 4	Hardware assembly and setup
Week 5	Hardware tuning, troubleshooting, trials and evaluation

3.5 Financial Plan and Costs

We bought most of the components from a local online shop, Techshopbd. We are five members in the group, and we funded the project ourselves, sharing the cost equally. The itemised cost of the main hardware is given in Table 5; with delivery and sourcing charges, the total project cost was approximately BDT 3,000, or about BDT 600 per member.

Table 5: Itemised cost of the main hardware

No.	Component	Price (BDT)
1	Arduino Uno R3	619.15
2	DC motor driver shield	230.03
3	Complete robot chassis (Tee01, red)	850.73
4	Servo motor SG90	219.01
5	Sonar sensor (HC-SR04)	109.33
6	4 × AA battery holder with jack	65.00
7	Female-to-female jumper wire (single)	54.40
8	Male-to-male jumper wire (single)	54.40
9	Male-to-female jumper wire (single)	51.20
10	DIP switch, 1 position	14.90
11	Double-sided tape, 0.5 in	89.96
12	Rechargeable battery unit	413.23
	Total	**2,771.34**

Because we had to buy every component individually in small quantities, the prototype cost is relatively high. If the robot were produced commercially, bulk purchasing and direct import would reduce the price of each component significantly. We estimate that a full-featured production version, including the UV, PIR and gas-sensing modules, can be built for no more than BDT 5,000.

Chapter 4: Methodology

4.1 Related Work

The development of autonomous devices that perform several tasks is an active area of research and commercial development. The most relevant existing examples are:

- **iRobot Roomba:** a robotic vacuum cleaner that uses sensors to navigate around a room and avoid obstacles, and a dirt-detection sensor to find areas that need extra cleaning.
- **LG Hom-Bot:** an autonomous vacuum cleaner that uses a camera, ultrasonic sensors and infrared sensors to map and navigate a room.
- **UV disinfection robots:** commercial and research robots, such as those used in hospitals, carry UV-C lamps on mobile bases to disinfect rooms [1], [2], [5], [8], [9]. These are effective but costly, and they do not clean floors.
- **Arduino obstacle-avoiding robots:** many academic prototypes use ultrasonic or IR sensors with an Arduino to avoid obstacles [4], [7], [10]; others add floor-cleaning functions [11], [12] or gas and fire detection with SMS alerts [13].

Table 6: Comparison of this project with related devices

Device	Obstacle avoidance	Floor cleaning	UV disinfection	Fire / gas alarm
Robotic vacuum (e.g. Roomba, Hom-Bot)	Yes	Yes	No	No
UV disinfection robot	Yes	No	Yes	No
Fixed fire / gas detector	No	No	No	Yes
This project	Yes	Yes	Yes	Yes

This project builds on existing work and aims to bring these capabilities together in one low-cost device that vacuum-cleans, disinfects, avoids obstacles and warns of fire or gas hazards.

4.2 Technical Approach

The technical approach is to build an automated system around an Arduino microcontroller. Various sensors respond to changes in the device's surroundings and enable it to perform several tasks. The main elements are:

- **Microcontroller:** the brain of the system. It receives signals from the sensors and uses them to activate the appropriate components.
- **Sensors:** an ultrasonic sensor for obstacles, a PIR sensor for human motion, and temperature and gas sensors for hazards.

- **Vacuum-cleaning mechanism:** a motor and fan with a suction opening, powered reliably from the main supply.
- **Fire and gas alarm:** the controller activates a buzzer and warning LED when a sensor threshold is exceeded.
- **Obstacle-avoidance mechanism:** when an obstacle is detected, the controller stops the wheel motors and changes direction.
- **Power supply:** a rechargeable battery pack, regulated to 5 V for the logic circuits.

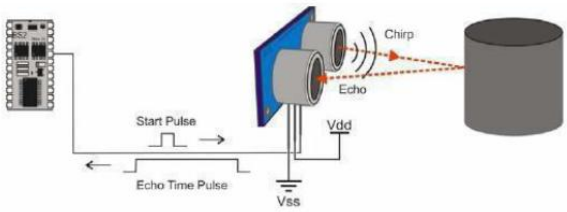


Figure 10: Working principle of the ultrasonic sensor: a 10 µs trigger pulse starts an 8-cycle burst, and the echo pulse width is proportional to the distance.

4.2.1 Obstacle avoidance

The vehicle detects objects with an ultrasonic sensor. The system started with a single sensor, but a vehicle has blind spots on its left and right, so the sensor is mounted on a servo that sweeps both sides, limiting the chance of collision. The robot is designed to treat anything inside a particular range as an obstacle, after which it avoids it and changes direction.

The sensor emits an ultrasonic pulse; the pulse is reflected by a nearby object and returns as an echo. The Arduino measures the time between the pulse and its echo and computes the distance using the speed of sound (about 340 m/s). In the code, the distance in centimetres is the echo pulse width in microseconds divided by 58.82 (the round trip at 340 m/s). When the measured distance is 30 cm or less, the controller stops the motors, turns the servo to the right and then to the left to measure the distance on each side, and finally turns the vehicle towards the side with the larger distance. If no obstacle is detected, the vehicle moves forward.

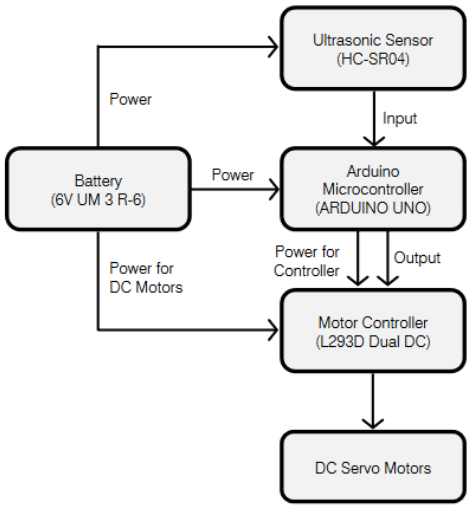


Figure 11: Block diagram of the obstacle-avoidance module: battery, ultrasonic sensor (HC-SR04), Arduino Uno, motor controller (L293D) and motors.

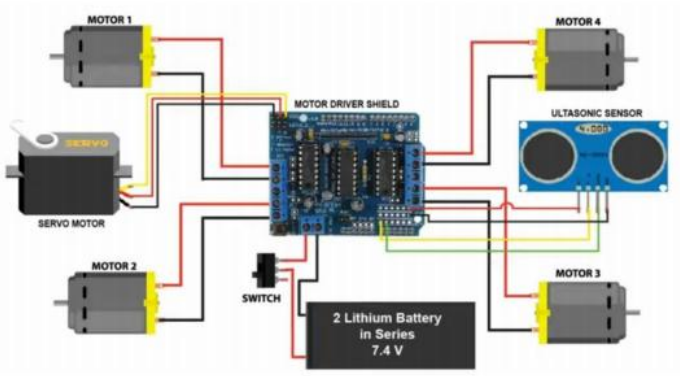


Figure 12: Wiring of the motor-driver shield with the DC motors, servo motor, ultrasonic sensor, switch and 7.4 V battery.

For obstacle detection many sensors are available, the popular ones being IR and ultrasonic sensors. We use the ultrasonic sensor, which continuously emits high-frequency pulses; when an object is present, the pulses are reflected back to the sensor.

4.2.2 Human-motion detection and UV switching

A PIR sensor detects human motion in close proximity (about 25 cm in our tests) and sends a digital HIGH signal to the Arduino, which then turns the UV LEDs off. The LEDs are turned on again once no human motion is detected within that range.

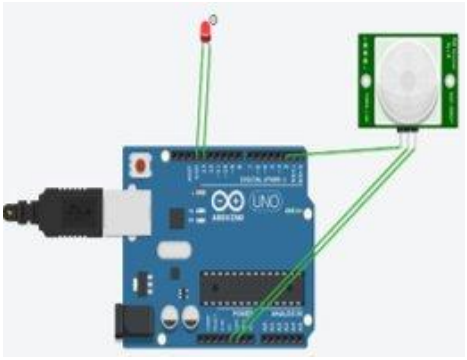


Figure 13: Arduino Uno connected to the PIR sensor and LED.

4.2.3 Fire and gas alarm

Fire alarm systems are crucial for the safety of buildings and factories. They typically use a group of sensors that monitor for flames, gas leaks or fire and trigger an alarm if one is detected. Common detection methods include infrared flame sensors, which use a photodiode to detect the infrared light emitted by a fire, and gas sensors, which change their output with the concentration of a gas. In our design, the temperature sensor (analog input A1) and the gas sensor (analog input A0) are read by the Arduino. When the temperature reaches 80 °C, a warning LED lights up, and when the gas reading reaches the threshold of 100, the buzzer sounds. The sensor values are also printed over the serial port once per second. These thresholds can be adjusted for the environment.

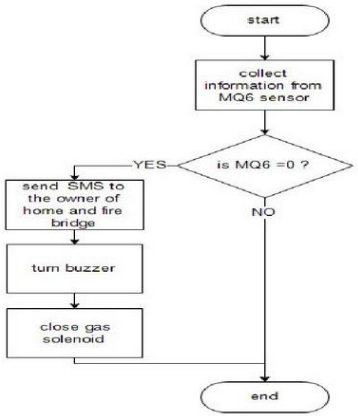


Figure 14: Reference gas-leak detection flowchart (adapted from [13]); SMS notification can be added with a GSM module.

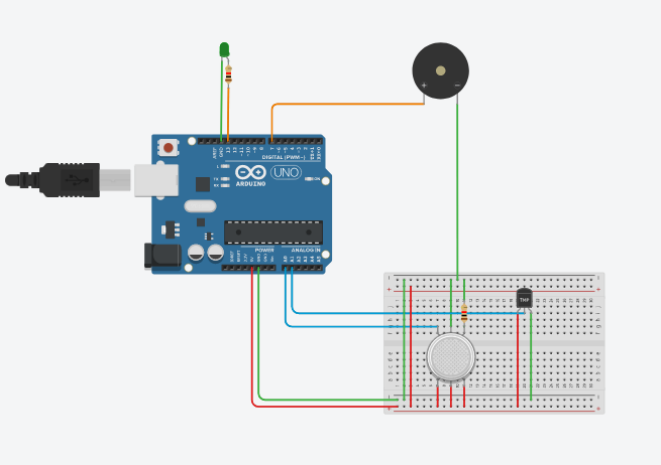


Figure 15: Arduino connected to a buzzer and indicator for the alarm circuit.

4.2.4 Vacuum cleaning

The vacuum module is based on a small portable vacuum cleaner. Such a cleaner has three main parts: a chamber that holds the dust, a middle section that contains the motor and fan, and a battery compartment. Its DC motor normally runs from 3 V (two 1.5 V AA cells) through a simple switch. In our design the original battery is removed, unnecessary parts are discarded and only the two-wire motor and fan are kept, powered from the robot's regulated supply.

To avoid falling down stairs, an IR proximity sensor can be mounted under the robot. It consists of an IR LED and a photodiode; the output changes when the emitted light is reflected from the floor, so a sudden change indicates an edge and the robot can stop and turn back. The three ultrasonic sensors (left, centre and right) share a common ground and 5 V supply, with their trigger and echo pins connected to Arduino pins. The L293D enable pins and driver supply are connected to 5 V, since the motors are 5 V types. A 7.4 V Li-ion battery supplies the system, and an LM7805 regulator provides the 5 V rail for the Arduino, ultrasonic sensors, motor driver and motors.

For the compact version, the circuit is soldered on a small perfboard using headers for an Arduino Pro Mini. The body is a round robot cut from MDF (chosen because it is hard and water-repellent), with a circular opening for the vacuum unit, two motors fixed by brackets, a castor wheel, the ultrasonic sensors on the left, centre and right, the IR module under the base, and an external power switch.

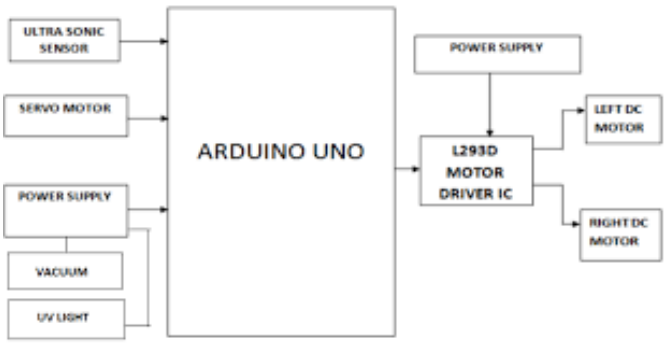


Figure 16: Block diagram of the complete system, including the vacuum cleaner and UV light.

4.3 Circuit Design and Block Diagrams

The block diagram below shows the system without the vacuum module. The Arduino is connected to the motor driver, which controls the motors running the wheels (left front, left back, right front and right back motors are connected to terminals M1–M4 of the driver). The ultrasonic sensor has four pins (VCC, Trig, Echo, GND); Trig works as an output and Echo as an input. The servo motor rotates the ultrasonic sensor. The power connection is a Li-ion battery pack in series (7.4 V) connected to the power board of the Arduino.

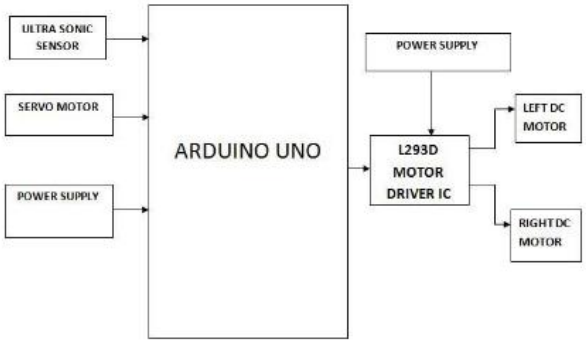


Figure 17: System block diagram (without the vacuum module).

The PIR motion sensor has three pins: VCC, OUT and GND. VCC and GND are connected to the 5 V and ground pins of the Arduino, and the output pin is connected to pin 2. The positive end of the LED is connected to pin 13 and the negative end to ground.

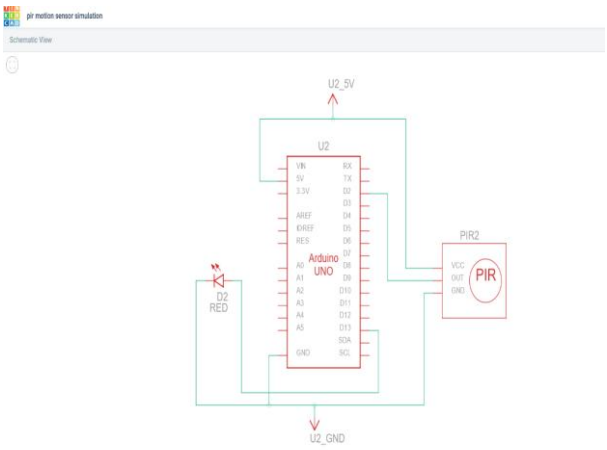


Figure 18: Tinkercad schematic of the PIR sensor and LED circuit.

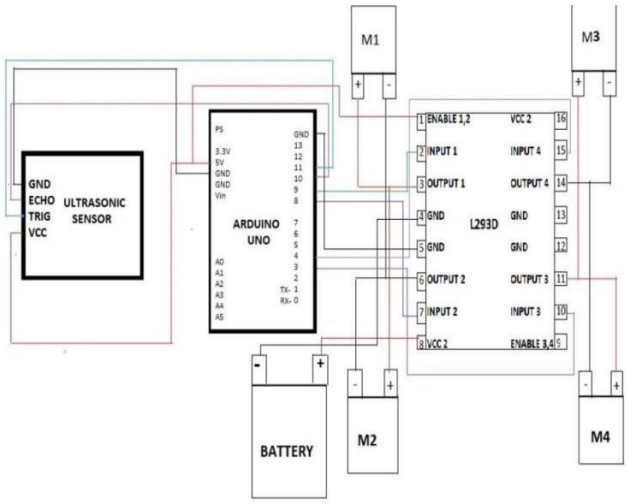


Figure 19: Circuit diagram of the Arduino, L293D motor driver, ultrasonic sensor and motors.

4.4 Simulation

Before assembling the hardware, the circuits were simulated in Tinkercad. The steps followed were:

1. **Select components:** from the Tinkercad library, choose parts similar to the ones to be used, such as an Arduino board, motors, LEDs, ultrasonic, PIR and gas sensors, and a power supply.
2. **Assemble the circuit:** drag the components into the workspace and connect them with virtual wires that match the intended physical layout.
3. **Configure properties:** set component properties such as the sensor trigger distance, the motor voltage and the gas-sensor sensitivity.
4. **Program the microcontroller:** upload the code for motor control, obstacle detection, motion detection and the alarm logic.

5. **Examine the findings:** run the simulation and check the motors, the obstacle detection and the alarm triggering. Identify problems and areas that need improvement.
6. **Iterate and refine:** adjust the circuit, component properties or program logic based on the results and repeat the simulation.
7. **Document:** keep screenshots and notes of key observations and conclusions.

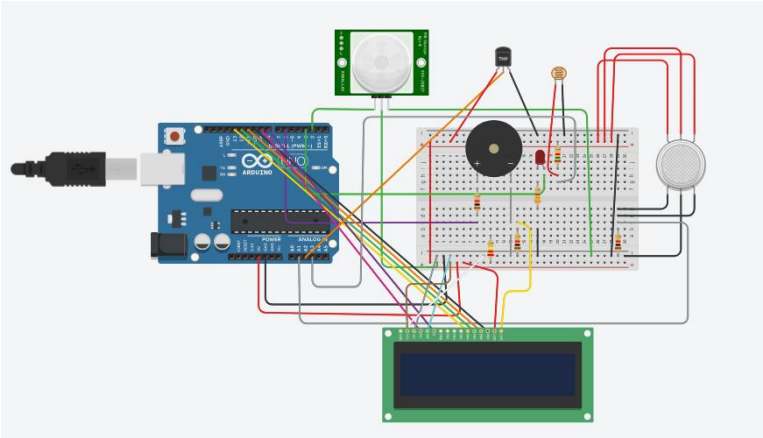


Figure 20: Tinkercad simulation of the PIR sensor, gas sensor, buzzer and LCD with an Arduino Uno.

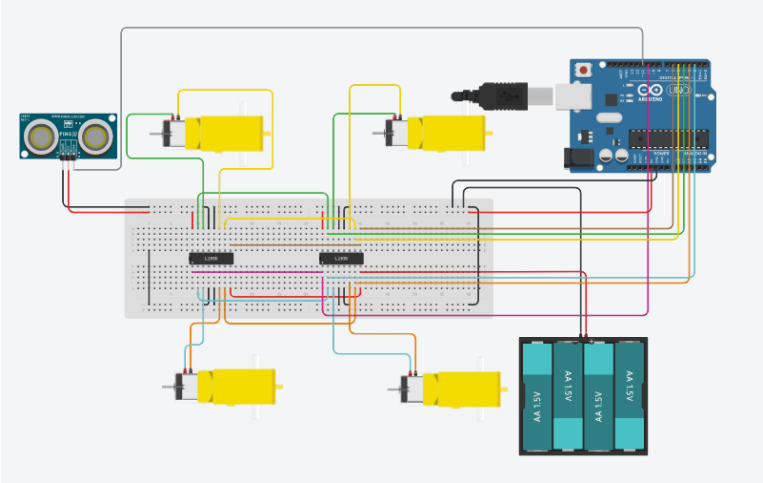


Figure 21: Tinkercad simulation of the obstacle-avoiding vehicle with ultrasonic sensors and motors.

4.5 Coding and Implementation

The control program is written in C++ using the Arduino IDE. It uses the AFMotor library to drive the four DC motors (LF, LB, RF, RB) through the motor shield and the Servo library for

the sensor servo. The complete listings are given in Appendix C. The obstacle-avoidance logic is summarised in the following flow chart and steps.

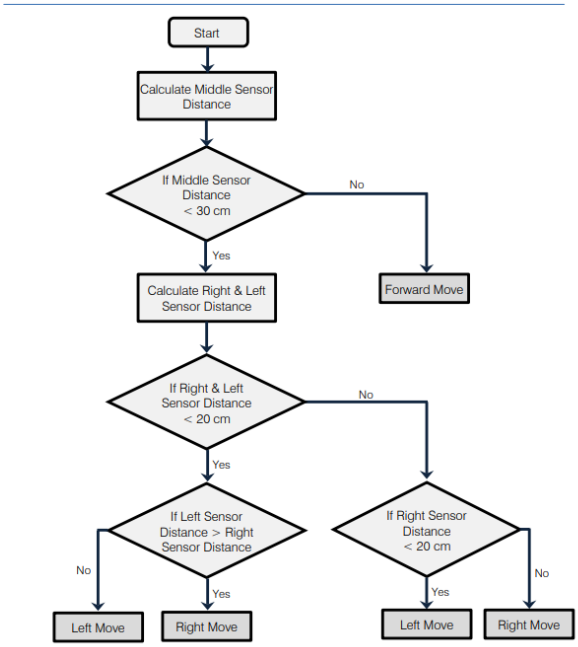


Figure 22: Flow chart of the obstacle-avoidance decision logic.

1. Initialise the servo on pin 10, the trigger pin (A0) and echo pin (A1), and set all four motors to speed 200.
2. Measure the front distance by sending a 10 µs trigger pulse and timing the echo.
3. If the distance is greater than 30 cm, run all motors forward and repeat from step 2.
4. If the distance is 30 cm or less, stop the motors, turn the servo to the right (5°) and measure the right distance, turn it to the left (180°) and measure the left distance, and return it to the centre (90°).
5. If the left distance is greater than the right distance, reverse briefly, stop, and turn left for 500 ms; otherwise reverse briefly, stop, and turn right for 500 ms.
6. Resume forward motion and repeat.

The motion-detection program reads the PIR output on pin 2 and switches the UV LED connected to pin 13 off when motion is detected, switching it on again when the output returns to LOW. The gas and temperature program reads analog inputs A0 and A1, turns on the warning LED when the temperature reaches 80 °C and the buzzer when the gas reading reaches 100, and prints both values to the serial monitor every second.

Chapter 5: Project Summary

5.1 Results and Discussion

In our tests the system detected obstacles and prevented collisions in about 97% of trials. Occasionally, when an obstacle appears suddenly in front of the device, the system fails to stop in time. There are two reasons: (i) the device is moving and accelerating constantly, and (ii) the controller needs a short time to process the sensor data after detecting the obstacle. Combined with the fact that our demonstration car has no dedicated braking system, so that stopping takes longer, these two factors cause the car to hit a suddenly appearing obstacle. Apart from this issue, the system performs consistently.

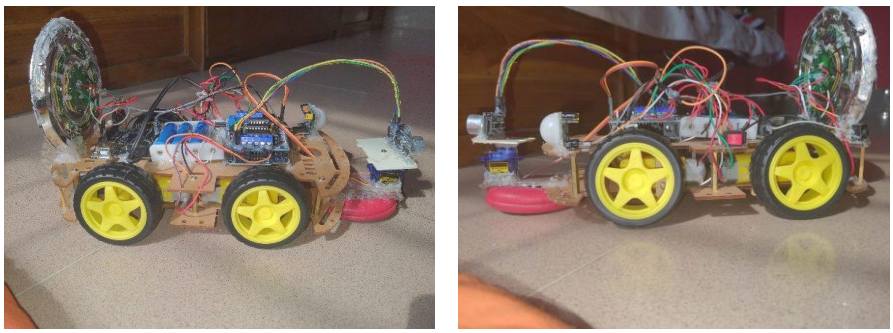


Figure 23: Four-wheel prototype vehicle, two views.

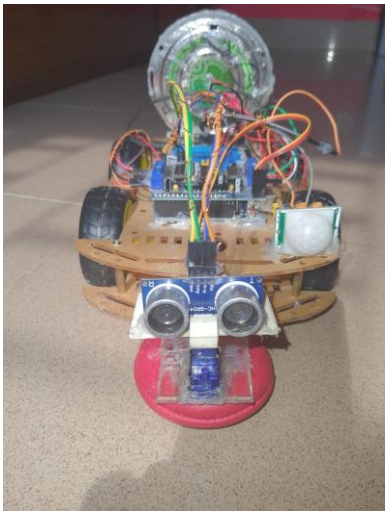


Figure 24: Compact round prototype with the ultrasonic sensor and PIR sensor.

Detection and avoidance of obstacles: the simulation and hardware results show how well the ultrasonic sensor detects obstacles and triggers the response. The device identified obstacles accurately and reacted quickly; the limitations noted above are the main points for improvement. Despite these limitations we are satisfied with the results.

Table 7: Implementation status of the modules

Module	Status	Evidence
Obstacle avoidance	Built and tested on the prototype	Photographs, Tinkercad simulation, code in Appendix C; about 97% collision-free trials
PIR-controlled UV LEDs	Built and tested	PIR/LED circuit, code in Appendix C, demonstration
Temperature and gas alarm	Coded and simulated	Tinkercad simulation and code in Appendix C
Vacuum cleaning	Designed	Mechanism and integration described in Section 4.2.4

5.2 Feasibility Study

We analysed whether the solution is feasible in cost, technical and operational terms.

Cost: the obstacle-avoidance kit that is added to an already-built car (Arduino Uno, motor-driver shield, servo motor and sonar sensor) costs only about BDT 1,180 (BDT 1,177.52 from Table 5). The safety and protection it provides are far more valuable than this cost.

Technical: the system is highly modular, so any part can be changed or upgraded easily if necessary. Almost every risk it involves has an easy and effective solution.

Operational: the system is efficient, operating successfully about 97% of the time.

Considering these factors, the system can be declared feasible.

5.3 Problems Faced and Troubleshooting

Although the concept and design appeared sound, several problems arose during implementation. We tried to solve every one of them (Table 8).

Table 8: Problems faced and solutions

Problem	Cause / observation	Solution
Implementation	The wiring diagram collected from the internet did not work properly when we applied it directly to our hardware.	We modified the diagram slightly and used two wheels instead of four.
Motor driver	The motor driver stopped working during testing.	We replaced it with a new motor driver.
Power supply	The Arduino code and the servo motor worked properly, but the two DC motors connected to the wheels did not run, because the battery supplied too little power.	We increased the battery supply (a 16 V pack was used) and the motors and wheels then started working.

5.4 Limitations

The device has certain limitations that need to be considered:

- **Limited functionality:** it performs only the tasks it is programmed for and may not adapt to new situations.
- **Environmental limitations:** performance may be affected by light level, surface texture or temperature.
- **Obstacle detection:** ultrasonic sensors may not detect soft, thin or angled obstacles or accurately estimate their size and shape.
- **Human detection:** PIR sensors respond only to motion, so a person who stays completely still may not be detected. UV exposure risk therefore remains, and the device should not be run while people are in the room.
- **UV effectiveness:** the disinfection effect depends on UV wavelength, intensity, distance and exposure time, and was not measured quantitatively in this project.
- **Power:** run time is limited by the battery and requires frequent recharging.
- **Reliability and durability:** hobby-grade components may malfunction and need regular maintenance.
- **Cost and regulation:** production cost may limit access for some users, and the device may be subject to regulatory requirements depending on location and use.

5.5 Future Development

We have several plans for improving the system. We will replace the Arduino Uno with a Raspberry Pi and optimise the code to reduce processing time. We will also add a rear sensor to detect obstacles behind the device. Further improvements include:

A) **Improved sensors:** more advanced sensors such as lidar or radar to improve obstacle detection.

B) **Artificial intelligence:** AI algorithms to improve decision making and let the device learn new environments.

C) **Communication:** a communication module to link the device to smartphones or smart-home systems, including SMS or app alerts for fire and gas events.

D) **Voice control:** voice recognition and natural language processing to respond to spoken commands.

E) **Energy efficiency:** more efficient components and optimised software to extend run time.

F) **Modular design:** a modular structure so users can customise functions for different environments.

G) **Durability:** more robust materials and rigorous testing.

H) **User interface:** a more user-friendly interface to interact with the device.

5.6 Project Demonstration Review

After the project was completed the robot started working. We noticed that it detected obstacles of all types with its sonar sensor. The robot moved by battery power on its wheels, and the servo motor swept continuously; whenever the sonar sensor detected an obstacle in front, the robot changed its direction and moved towards the side with more space. The system worked as expected, and the modules demonstrated worked almost flawlessly. The ultrasonic sensor detects obstacles on the robot's path and signals the Arduino, which, through the motor driver, redirects the robot in a different direction to avoid them.

The system is designed to switch off the UV lights automatically when a person is in close proximity. PIR sensors detect human presence by sensing changes in infrared radiation. Because of this, they cannot reliably detect a stationary person, but nearby human motion is detected clearly, which is enough to switch the UV LEDs off. The device moved around an enclosed space and, with the UV LEDs on, was able to expose surfaces to UV light. Disinfection effectiveness against specific pathogens (including the virus that causes COVID-19) depends on the UV wavelength and dose and was not measured in this project.

To recap, we have built an obstacle-avoiding vehicle controlled by an Arduino that needs no human intervention while moving through an unfamiliar space. The project is designed to be a

robust, efficient and cost-effective autonomous device suitable for clinics, hospitals, offices, educational institutions and similar places. Its temperature and gas sensing adds an alarm function to reduce the risk of fire.

5.7 Lessons Learnt

The main lessons from developing and implementing the project are:

- **Interdisciplinary collaboration:** the project needed electronics, programming, mechanical and safety knowledge, which shows the value of teamwork across disciplines.
- **Testing and validation:** each component must be tested thoroughly. Regular testing and simulation throughout development helped us find problems early, such as the motor-driver and power-supply faults.
- **Power budgeting:** motors, UV LEDs and the vacuum fan draw much more current than the logic circuits, so the power supply must be planned early.
- **Safety engineering:** devices that interact with people must include safety features such as automatic shut-off (as with the PIR cut-off for UV) and shielding from harmful radiation.
- **Iterative design:** building and testing prototypes, and adapting published designs to the hardware we had, helped refine the design.
- **User needs:** the design should consider the needs and preferences of users so that the device is practical, easy to use and valuable.

5.8 Conclusion

Robotics is an interdisciplinary area of engineering and computer science concerned with the design, manufacture, operation and use of robots. It aims to create devices that support and help humans, and the use of robots is increasing day by day.

This project designed and built a multi-tasking autonomous device based on the Arduino platform. The device detects and avoids obstacles on its path using a servo-mounted ultrasonic sensor, switches its UV LEDs off when it detects human motion, and monitors temperature and gas to raise an alarm; a vacuum-cleaning module extends its usefulness for household cleaning. It operates without human intervention in an unfamiliar indoor environment and is suitable for clinics, hospitals, offices, educational institutions and banks. The prototype cost about BDT 3,000, and a full-featured version is estimated at no more than BDT 5,000.

The project also gives students and beginners a basic guide to embedded systems, sensors, microcontrollers and robot building with Arduino. Although the result is modest, it demonstrates that a low-cost robot can combine navigation, disinfection and safety monitoring, and it provides a base for the improvements described in Section 5.5.

Appendix A: Contribution of Individual Members

Table 9: Contribution of each member by project phase

Member	Requirement analysis	Planning	System analysis	Design	Implementation	Testing & development
Jubair Ibn Omar Hamza	20%	40%	20%	30%	30%	25%
Abdullah Al Mahfuz	20%	10%	20%	10%	15%	15%
Sartaj Mahmud	20%	25%	20%	30%	30%	25%
Nazifa Syed Aasfi	20%	15%	20%	20%	15%	20%
Shakil Khan	20%	10%	20%	10%	10%	15%
Total	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**

Appendix B: References

[1] Frontiers in Robotics and AI, “Exploring the Applicability of Robot-Assisted UV Disinfection in Radiology,” [Online]. Available: https://www.frontiersin.org/articles/10.3389/frobt.2020.590306/full

[2] “Robot Development and Path Planning for Indoor Ultraviolet Light Disinfection,” [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9561405

[3] A. Begić, “Application of Service Robots for Disinfection in Medical Institutions,” [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-319-71321-2_89

[4] M. Ghaleb, “Design of an Obstacle-avoiding Robot Car Based on Arduino Microcontroller,” [Online]. Available: https://www.researchgate.net/publication/329042229_Design_of_an_Obstacle-avoiding_Robot_Car_Based_on_Arduino_Microcontroller

[5] “Development of a UVC-based disinfection robot,” Industrial Robot, [Online]. Available: https://www.emerald.com/insight/content/doi/10.1108/IR-10-2021-0227/full/html

[6] K. McAllister, “Meet the robots working to disinfect your office,” Protocol, [Online]. Available: https://www.protocol.com/fetch-robotics-melonee-wise-interview

[7] “Obstacle-avoiding robot with IR and PIR motion sensors,” IOP Conf. Ser.: Mater. Sci. Eng., [Online]. Available: https://iopscience.iop.org/article/10.1088/1757-899X/152/1/012064/meta

[8] NIH, “Ultraviolet disinfection robots to improve hospital cleaning: Real promise or just a gimmick?,” [Online]. Available: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7879404/

[9] “An Ultra-violet sterilization robot for disinfection,” [Online]. Available: https://ieeexplore.ieee.org/abstract/document/8802528

[10] “Design of an Indicative Featured and Speed Controlled Obstacle Avoiding Robot,” [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9068018

[11] “Arduino Based Dry and Wet Automatic Floor Cleaner,” ResearchGate, Apr. 2020. [Online]. Available: https://www.researchgate.net/publication/341243272_ARDUINO_BASED_DRY_WET_AUTOMATIC_FLOOR_CLEANER

[12] “Development of a vacuum cleaner robot,” ResearchGate, Nov. 2018. [Online]. Available: https://www.researchgate.net/publication/328948615_Development_of_a_vacuum_cleaner_robot

[13] “An SMS Based Fire Alarm & Gas Leakage Detector Using Arduino and GSM Module,” Studocu, [Online]. Available: https://www.studocu.com/row/document/simad-university/java-programming-ii/an-sms-based-fire-alarm-gas-leakage-detector-using-arduino-and-gsm-module/22857658

[14] International Research Group, “Design and Implementation of IoT Based Smart Surveillance and Intelligent Monitoring System Using Raspberry Pi,” [Online]. Available: https://www.academia.edu/36992924/Design_and_Implementation_of_IOT_Based_Smart_Surveillance_and_Intelligent_Monitoring_System_Using_Raspberry_Pi

[15] “IoT based warfare car robot using sensor network connectivity,” ACM Digital Library, [Online]. Available: https://dl.acm.org/doi/10.1145/3231053.3231121

[16] “Remote Control Antennas,” Remote Control & Technology, [Online]. Available: https://remotecontroltech.com/accessories/remote-control-antennas

Appendix C: Codes

C.1 Human motion detection (PIR) code

```
int ledPin = 13;                // choose the pin for the LED
int inputPin = 2;               // choose the input pin (for PIR sensor)
int pirState = LOW;             // we start, assuming no motion detected
int val = 0;                    // variable for reading the pin status
void setup() {
  pinMode(ledPin, OUTPUT);      // declare LED as output
  pinMode(inputPin, INPUT);     // declare sensor as input
  Serial.begin(9600);
}
void loop(){
  val = digitalRead(inputPin);  // read input value
  if (val == HIGH)              // check if the input is HIGH
  {
    digitalWrite(ledPin, LOW);  // turn LED ON
    if (pirState == LOW)
  {
      Serial.println("Motion detected!"); // print on output change
      pirState = HIGH;
    }
  }
  else
  {
    digitalWrite(ledPin, HIGH); // turn LED OFF
    if (pirState == HIGH)
  }
```

PIR motion-detection program

C.2 Gas and temperature detection code

```
float temp;
float vout;
float vout1;
int LED = 13;
int gasSensor;
int piezo = 7;
void setup()
{
pinMode(A0,INPUT);
pinMode(A1, INPUT);
pinMode(LED,OUTPUT);
pinMode(piezo,OUTPUT);
Serial.begin(9600);
}
void loop()
{
```

Gas and temperature detection program (part 1)

```
void loop()
{
vout=analogRead(A1);
vout1=(vout/1023)*5000;
temp=(vout1-500)/10;
gasSensor=analogRead(A0);
if (temp>=80)
{
digitalWrite(LED,HIGH);
}
else
{
digitalWrite(LED,LOW);
}
if (gasSensor>=100)
{
digitalWrite(piezo,HIGH);
}
```

Gas and temperature detection program (part 2)

```
  if (gasSensor>=100)
  {
  digitalWrite(piezo,HIGH);
  }
  else
  {
  digitalWrite(piezo,LOW);
  }
  Serial.print("in DegreeC= ");
  Serial.print(" ");
  Serial.print(temp);
  Serial.print("\t");
  Serial.print("GasSensor= ");
  Serial.print(" ");
  Serial.print(gasSensor);
  Serial.println();
  delay(1000);
  }
```

Gas and temperature detection program (part 3)

C.3 Obstacle-avoidance code

```
#include <AFMotor.h>
#include <Servo.h>
Servo servo;
AF_DCMotor LF(1);
AF_DCMotor LB(2);
AF_DCMotor RF(3);
AF_DCMotor RB(4);
#define trigPin  A0
#define echoPin  A1
void setup()
  {
    servo.attach(10);
    pinMode(trigPin, OUTPUT);          // Declare "trigPin" as "Output Pin".
    pinMode(echoPin, INPUT);           // Declare "echoPin" as "Input Pin".
    LF.setSpeed(200);
    LB.setSpeed(200);
    RF.setSpeed(200);
    RB.setSpeed(200);
  }
void loop()
  {
    float distance = 0.00;
    float RightDistance = 0.00;


 float LeftDistance = 0.00;
    distance = search();
    if(distance <= 30) // If obstracle found in 40 CM.
```

Obstacle-avoidance program (part 1)

```
   }
 void loop()
   {
     float distance = 0.00;
     float RightDistance = 0.00;


  float LeftDistance = 0.00;
     distance = search();
     if(distance <= 30) // If obstracle found in 40 CM.
       {
           RobotStop(); //Robot Stop
           delay(100);
           servo.write(5);
           delay(300);
           RightDistance = search();
           delay(100);
           servo.write(180);
           delay(300);
           LeftDistance = search();
           delay(100);
           servo.write(90);
           delay(300);
          if(LeftDistance > RightDistance)
            {
              RobotBackward();
              delay(100);
              RobotStop();
              delay(100);
              RobotLeft();
```

Obstacle-avoidance program (part 2)

```
                {
                   RobotBackward();
                   delay(100);
                   RobotStop();
                   delay(100);
                   RobotLeft();
                   delay(500);
                   RobotStop();
                   delay(100);
                }
               else
                 {
                   RobotBackward();

                   delay(100);
                   RobotStop();
                   delay(100);
                   RobotRight();
                   delay(500);
                   RobotStop();
                   delay(100);
                 }
             }
           else
           {
             RobotForward();
           }
       }
       float search(void)
         {
```

Obstacle-avoidance program (part 3)

```
float search(void)
  {
    float duration = 0.00;             // Float type variable declaration
    float CM = 0.00;
    digitalWrite(trigPin, LOW);        // Trig_pin output as OV (Logic Low-Level)
    delayMicroseconds(2);              // Delay for 2 us
    //Send 10us High Pulse to Ultra-Sonic Sonar Sensor "trigPin"
    digitalWrite(trigPin, HIGH);       // Trig_pin output as 5V (Logic High-Level)
    delayMicroseconds(10);             // Delay for 10 us
    digitalWrite(trigPin, LOW);        // Trig_pin output as OV (Logic Low-Level)
    duration = pulseIn(echoPin, HIGH);
    CM = (duration / 58.82); //Convert distance into CM.
    return CM;
  }
void RobotForward()

{
  LF.run(FORWARD);
  RF.run(FORWARD);
  LB.run(FORWARD);
  RB.run(FORWARD);
}
void RobotBackward()
{
  LF.run(BACKWARD);
  LB.run(BACKWARD);
  RF.run(BACKWARD);
  RB.run(BACKWARD);
}
```

Obstacle-avoidance program (part 4)

```
void RobotBackward()
{
  LF.run(BACKWARD);
  LB.run(BACKWARD);
  RF.run(BACKWARD);
  RB.run(BACKWARD);
}
void RobotLeft()
{
  LF.run(BACKWARD);
  LB.run(BACKWARD);
  RF.run(FORWARD);
  RB.run(FORWARD);
}
void RobotRight()
{
  LF.run(FORWARD);
  LB.run(FORWARD);
  RF.run(BACKWARD);
  RB.run(BACKWARD);
}
void RobotStop()
{
  LF.run(RELEASE);
  LB.run(RELEASE);

  RF.run(RELEASE);
  RB.run(RELEASE);
}
```

Obstacle-avoidance program (part 5)